

ResortHub Management System

Official Functional & Business Requirements Specification

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Submission Date: September 23, 2026

PREPARED FOR Engineering / Operations Manager	PROJECT ResortHub Luxury Hospitality Platform	VERSION 2.4.0 (Production Release)	SIGN-OFF STATUS READY FOR APPROVAL
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1 Executive Summary & Purpose

This document outlines the complete business, functional, and operational requirements for **ResortHub**, an all-in-one luxury hospitality SaaS platform. It combines a guest booking interface with enterprise operations across 5 organizational roles (Guests, Front Desk, Resort Managers, Accountants, and SaaS Co-Owners).

Primary Business Objectives:

- Deliver a **modern, luxury mobile/web discovery interface** for guests that maximizes conversion with interactive amenity filters, transparent pricing, and instant maps integration.
- Provide a **split-payment engine (35% advance deposit)** to lower initial booking friction while guaranteeing reservations.
- Automate contactless arrival with **digital encrypted QR check-in passes**.
- Provide comprehensive operational back-office tooling for housekeeping, inventory control, financial ledger reconciliation, and **multi-tenant SaaS portfolio management**.

2 User Personas & Permission Scope

Role	Target User	Functional Responsibility	Access Boundary
Customer	Resort guests, vacation travelers	Browse luxury catalog, apply amenity filters, book rooms, pay split deposit, view digital QR pass.	Public discovery & personal bookings
Incharge	Duty managers, front desk staff	Scan guest QR passes, execute check-in/out, dispatch housekeeping tasks, monitor room status grid.	Single property operations desk
Admin	General Managers, Property Owners	Set room tariffs, configure dynamic pricing, manage staff shifts, create promotional campaigns.	Full property administrative controls
Accountant	Finance & audit personnel	Audit split payments (advance vs. on-premise dues), reconcile F&B/spa billing, export tax & P&L reports.	Property financial records & ledgers
Super Admin	Platform Operators, Co-Owners	Monitor 21+ subscribed resorts, manage SaaS subscription tiers, onboard new resorts, track platform MRR.	Platform-wide multi-tenant controls

3 Core Functional Requirements (Submitted for Verification)

REQ-01: Luxury Guest Discovery & UI Experience

PRIORITY: HIGH

The guest interface must deliver a high-end luxury aesthetic utilizing a soft mint theme (#EDF6F2 Light / #0B1118 Dark) with emerald green accents (#10B981) rather than standard corporate blue styling.

Verification Criteria: Header displays greeting, live location badge ("Bengaluru, India"), profile access, and a one-click Light/Dark theme toggle. No blue background artifacts.

REQ-02: Interactive Amenities Filter Row

PRIORITY: HIGH

The home discovery screen must provide a horizontal scrolling amenity selector with live selection indicators (Pool, Free Wi-Fi, Spa, Restaurant, Beachfront, Gym, Bar, Pet-Friendly).

Verification Criteria: Selecting any amenity filter instantly filters the resort list in real time with active visual highlight badges and count displays.

REQ-03: Resort Details, Gallery Counter & Maps Redirection

PRIORITY: HIGH

The property details screen must showcase a multi-image carousel with real-time photo index indicators (e.g. "1 / 4"), key amenity cards, nightly pricing, and clickable location coordinates.

Verification Criteria: Clicking the resort address or map pin opens Google Maps in a new tab with exact GPS coordinates. Photo carousel tracks active index accurately.

3 Core Functional Requirements (Continued)

REQ-04: Split-Payment Booking Engine (35% Deposit)

PRIORITY: CRITICAL

Guests must be allowed to confirm reservations by paying an advance deposit of 35% online, with the remaining 65% balance scheduled for payment upon physical check-in.

Verification Criteria: Booking modal itemizes total charges, 35% advance payable immediately, and 65% balance due on arrival. Transaction records reflect split payment states accurately.

REQ-05: Digital Contactless QR Check-in Pass

PRIORITY: HIGH

Upon successful reservation, the system must generate a digital QR room pass containing encrypted reservation credentials for rapid front desk check-in.

Verification Criteria: Pass displays reservation number, guest name, check-in date, room tier, and a high-resolution scannable QR code.

REQ-06: Front Desk Operations & Housekeeping Dispatch (Incharge)

PRIORITY: HIGH

Duty managers require a unified operational dashboard to scan guest passes, transition room statuses (Available, Occupied, Cleaning, Maintenance), and dispatch housekeeping staff.

Verification Criteria: Front desk can perform 1-click check-ins, record guest service tabs, and track housekeeping cleaning SLAs.

REQ-07: Financial Ledger & Split Settlement Reconciliation (Accountant)

PRIORITY: HIGH

Accountants require real-time visibility into advance gateway collections vs. front desk cash/card collections, daily revenue, and tax summaries.

Verification Criteria: Automatic categorization into Room, Food & Beverage, Spa, and Ancillary ledgers with exportable financial reports.

REQ-08: Multi-Tenant Super Admin SaaS Management Portal

PRIORITY: CRITICAL

Super Admins and platform co-owners require a centralized dashboard to monitor all subscribed resort properties, track subscription tiers, and onboard new properties.

Verification Criteria: Telemetry cards displaying "21 Subscribed Resorts", "21 Active Subscriptions", Tier dropdown filter ("All Tiers", "Starter", "Pro", "Enterprise"), live search bar, and "+ Add Resort" creation flow.

REQ-09: Universal Light/Dark Theming & Browser Cache Control

PRIORITY: MEDIUM

Every screen (guest discovery and administrative modules) must seamlessly support both Light Mode and Dark Mode. Furthermore, web deployment must prevent stale service worker caching.

Verification Criteria: Instant toggle across all views; PWA caching bypassed so browser always renders fresh assets without manual hard refreshes.

4 Acceptance & Sign-off Checklist for Manager

- ✓ **Design Compliance:** Soft mint luxury theme applied; blue background completely removed.
- ✓ **Amenity Filtering:** Working horizontal toggle chips for Pool, Wi-Fi, Spa, Restaurant, etc.
- ✓ **Location & Navigation:** Clickable resort location opening live Google Maps coordinates in new tab.
- ✓ **Payment Flexibility:** 35% Advance deposit option enabled with transparent 65% balance due on arrival.
- ✓ **Contactless Check-in:** Scannable QR digital room pass generated upon booking.
- ✓ **Super Admin Portal:** Multi-tenant SaaS dashboard managing 21 subscribed resorts and subscription tiers.
- ✓ **Quality Assurance:** 0 static analysis errors (flutter analyze passed); unit tests verified (4/4 passed).

Prepared By: Frontend Engineering Lead

Signature: _____

Date: September 23, 2026

Approved By (Manager): _____

Signature: _____

Date: _____